University of Nottingham Malaysia

SCHOOL OF COMPUTER SCIENCE
A LEVEL 1 MODULE, SPRING SEMESTER 2021-2022

DATABASES AND INTERFACES (COMP 1044)

Time allowed: **TWO** Hour

An additional 30 minutes to cover IT issues and upload time

Answer ALL questions

INSTRUCTION FOR STUDENTS:

- The university's honour code applies to this individual short time assessment. The submitted answers **must be your own work** where **answers are attempted without any help from others**. University academic misconduct rules applies here in the case of plagiarism, collusion or false authorship.
- Write the name and student ID at the top of your answer sheets.
- Students are not allowed to ask questions during the short time assessment in accordance with the no shout-out policy.
- You may answer the questions in any order but write the question numbers correctly and clearly.
- At the end of the exam, **upload your** solutions and name the file as **StudentID.** Upload your solution file onto **Moodle**.
- Please ensure that you upload the correct files to the folder labelled as "Final examination submission folder" in Moodle. Request for resubmission will not be accepted after the submission.
- Students should stop writing the answers at the end of exam time. Answer scripts must be submitted to Moodle within **thirty minutes** after the end of exam. In case of any technical issues, the students need to contact the module convener in this period.

IMPORTANT: Lateness penalties will not be applied, and late submission will not be accepted.

Question 1 ERD [25 marks]

a) Signature Cabinet Sdn Bhd, a kitchen cabinet and wardrobe manufacturer company has contacted you to create a conceptual model by using the Entity–Relationship data model for a database that will meet the information needs for its supply chain management system. The Company Director gives you the following description of the supply chain operating environment.

The manufacturing company produces products. The following product information is stored: product name, product ID and quantity on hand. These products are made up of many components. Each component can be supplied by one or more suppliers. The following component information is kept: component ID, name, description, suppliers who supply them, and products in which they are used.

Create a logical ERD to show how you would track this information. Show entity names, primary keys, attributes for each entity, relationships between the entities and cardinality.

Assumptions

- A supplier can exist without providing components.
- A component does not have to be associated with a supplier.
- Not all components are used in products.
- A product cannot exist without components.

[16 marks]

b) Attributes are the properties of entities. Briefly explain the following type of attributes and give an example for each of the following:

i. Derived attribute
ii. Multivalued attribute
iii. Composite attribute

[9 marks]

Question 2 SQL [25 marks]

The tables below are used by a company to record data about its employees and departments.

Table name: STUDENT						
SID	**SNAME**	**GENDER**	**AGE**	**CITY**	**CONTACTNO**	**CNO**
20001	HELEN	F	18	CHERAS	60163247274	2001
20002	LEE	M	19	SEMENYIH	60123135298	1001
20003	ALEX	M	20	KAJANG	60143056237	3001
20004	TOM	M	20	CHERAS	60163533839	4001
20005	JAMES	M	19	SERDANG	60173284620	2001
20006	KIM	F	24	PUCHONG	60113796231	4001
20007	JOHN	M	21	KAJANG	60123637214	5001

Table name: COURSE		
CNO	**CNAME**	**FEE**
1001	ACCOUNTING	110000
2001	CIVIL ENGINEERING	230000
3001	PSYCHOLOGY	187000
4001	COMPUTER SCIENCE	120000
5001	EDUCATION	89000

a) Formulate SQL statements for the following queries:

i. Display the number of students who stay at Kajang. **[2 marks]**

ii. Increase the fee of all courses by 10%. **[2 marks]**

iii. Display all the student's name who study computer science. **[3 marks]**

iv. Display student ID that are not staying in Kajang and Semenyih. **[3 marks]**

v. Display the course fee that need to pay by the student (SID: 20005) **[3 marks]**

vi. Calculate the total amount of fees that will be paid by all the students. **[4 marks]**

vii. Display student ID and student name that have same course with the student (SID: 20004) **[4 marks]**

b) Differentiate the result produced by these two SQL statements:

- "***SELECT gender, age FROM Student;***"
- "***SELECT DISTINCT gender, age FROM Student;***"

[4 marks]

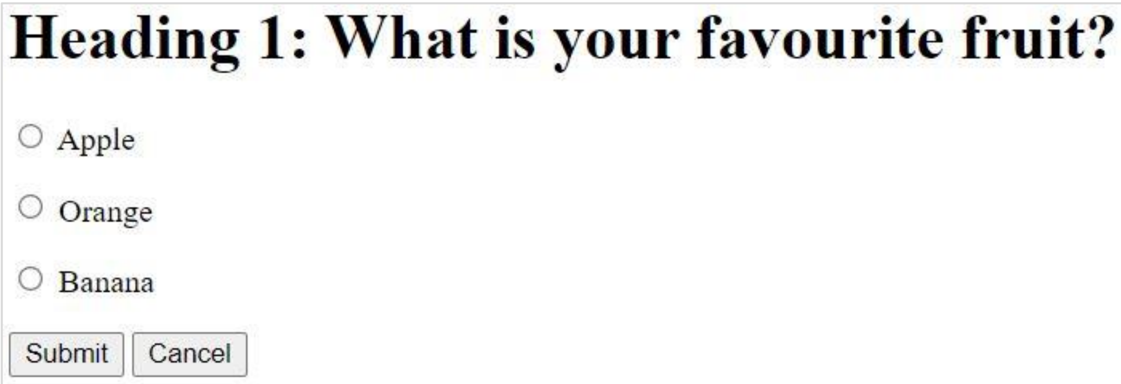
Question 3 HTML& CSS **[25 marks]**

a) A CSS selector selects the HTML element(s) you want to style. Briefly explain the difference between id selector and class selector and describe with example of CSS style rules for the following HTML snippet.

<p class="quote" id="p1">An apple a day keeps doctor away!</p>

[6 marks]

b) Write the HTML code that produces following output. The code should include all necessary elements such as HTML, HEAD and BODY.



[11 marks]

c) What is the difference between static and dynamic web pages? Why are dynamic web pages used? **[4 marks]**

d) <span> and <div> tags are useful for style specifications. Explain when to use <span> tag and <div> tag and provide justification with example.

[4 marks]

Question 4 PHP & Javascript [25 marks]

a) Consider the following fragment of JavaScript. Give the output that would be produced in the browser window by the pieces of code in parts (a) through (d) below.

i.
```
var i;
for (i = 2; i < 8; i += 2) {
        document.writeln(“The value of i is ”+i+“.”);
}
```

ii.
```
function output(option)
{
        return (option ? "yes" : "no");
}
var ans=false;
document.writeln(output(ans));
```

iii.
```
var grade='A';
var result=0;
switch(grade)
{
        case 'A':
                result+=4;
        case 'B':
                result+=3;
        case 'C':
                result+=2;
        default:
                result+=1;
}
document.writeln(result);
```

[7 marks]

b) Consider the following HTML page. Write JavaScript code to modify the given HTML in order to make the background change to red under the specified circumstances, by calling changeBG() through an event handler.

```
<html>
<head>
<title>A Basic Webpage</title>
</head>
<body >
<script>
        function changeBG() {
                document.bgColor = "red"
        }
</script>

<a href="link.html">A link</a>

</body>
</html>
```

i. Immediately when the page appears.
ii. If another window is opened on top of the current one.
iii. If the user clicks on the link.
iv. If the user moves the mouse over the link.

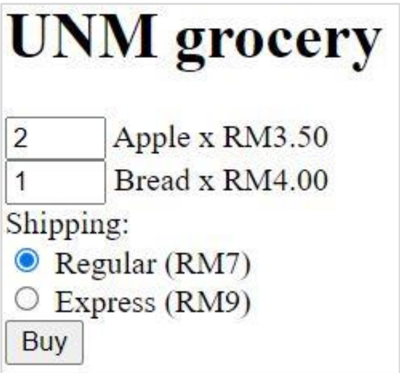
[8 marks]

c) Suppose the following HTML code exists for ordering grocery online:

```
<form action="output.php" method="post">
    <div><input id="input1" type="text" name="apple" size="2" /> Apple x RM3.50</div>
    <div><input id="input2" type="text" name="bread" size="2" /> Bread x RM4.00</div>
    <div>Shipping: <br />
    <label><input id="input3" type="radio" name="shipping" value="regular" /> Regular (RM7)</label> <br />
    <label><input id="input4" type="radio" name="shipping" value="express" /> Express (RM9)</label></div>
    <div><input id="input6" type="submit" value="Buy" /></div>
</form>
```

Your task is to write the PHP code that would be saved in output.php to process the grocery order form. You should display an h1 heading saying "Your order:" followed by total order cost. Print a "Thank you for your order!" message at the bottom of your output.

The screenshots below show example values typed into the form, along with your page's resulting appearance:



[10 marks]

END